

Introduction to Kubernetes

Instructor

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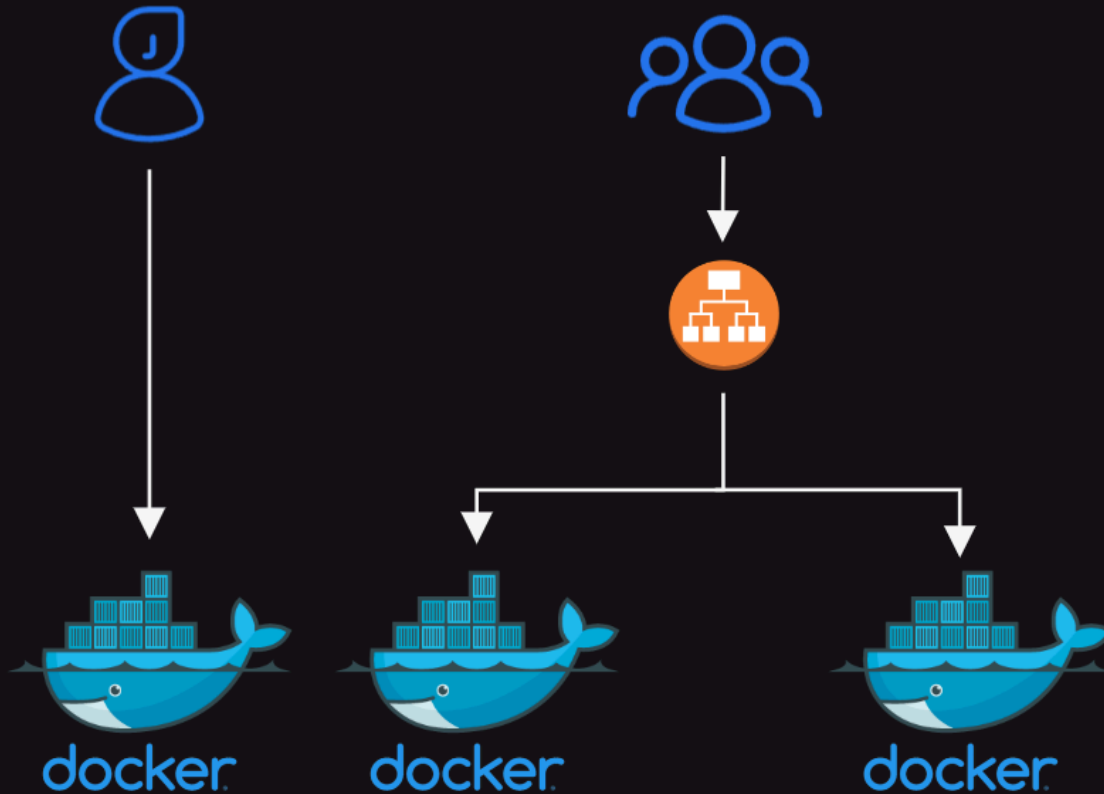
Ex-Head of Engineering - YC-backed startup

Certified AWS Cloud and Kubernetes Engineer

Open source contributor



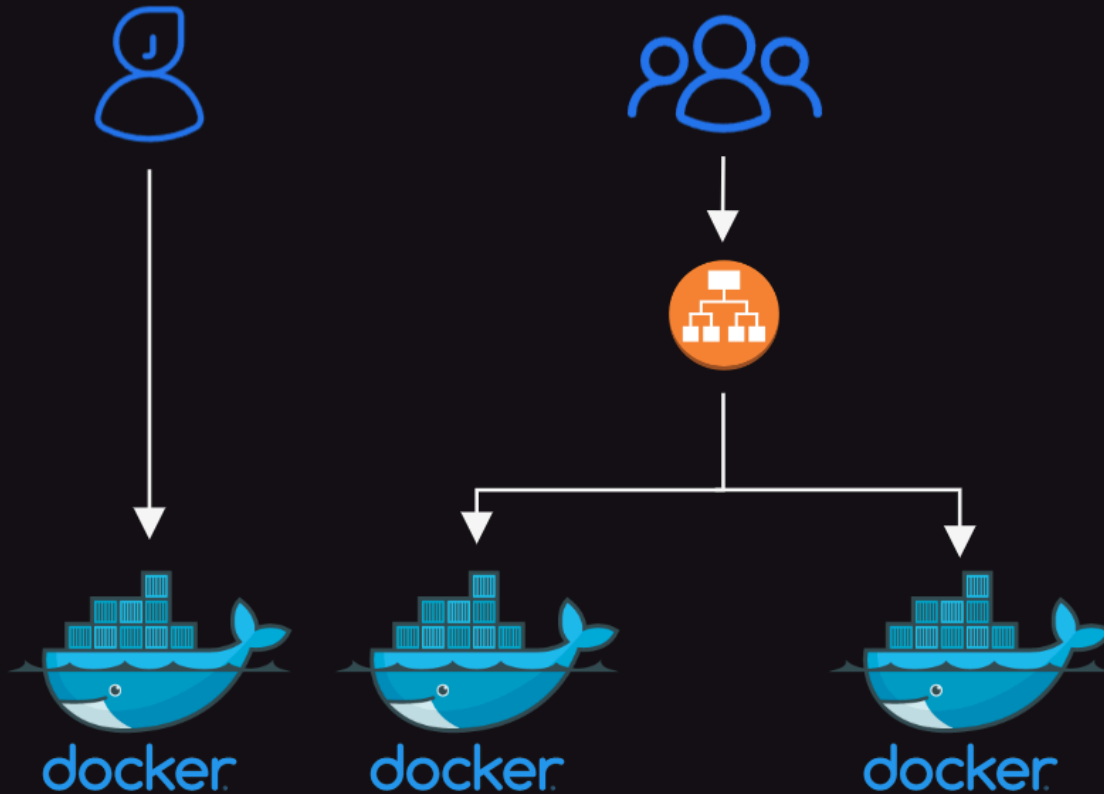
Managing Containers



As services grow, applications often shift to multiple microservices, leading to:

- **Manual Server Management:**
 - Increased complexity in managing servers.
- **EC2 Health Checks:**
 - Regular monitoring of machine health.
- **Volume Backup:**
 - Ensuring data safety through regular backups.

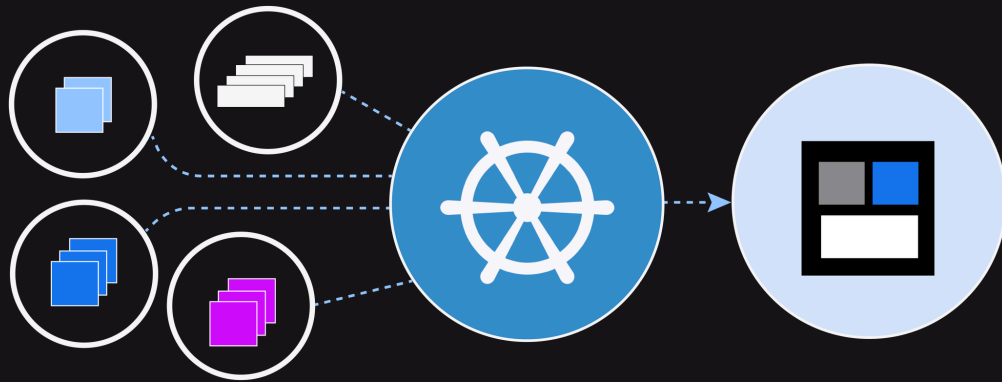
Managing Containers



Managing applications at scale presents several challenges:

- Reproducible and smooth deployment
- Secrets and environment variable management
- Careful resource allocation
- Logging, monitoring and auto-healing
- Microservice communication
- Auto-scaling and load-balancing
- Fine-grained application access to engineers

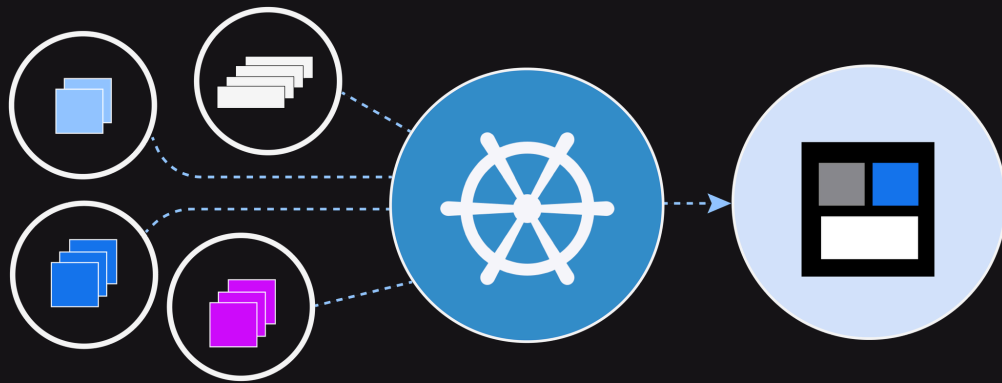
Kubernetes



Kubernetes or K8s, is an open-source platform designed to automate:

- **Deployment:**
 - Streamlines the process of deploying applications.
- **Scaling:**
 - Efficiently scales applications as needed.
- **Management:**
 - Facilitates the management of containerized applications.

Kubernetes



Key Features

- **Container Grouping:**
 - Organizes containers into logical units for easy management and discovery.
- **Proven Experience:**
 - Leverages 15 years of production workload experience at Google, enhanced by community-driven ideas and best practices.

Kubernetes Features



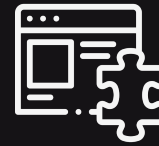
Fully manage application lifecycle



Auto scaling of containers and nodes



Easy microservice communication



Third party plugins



Role based access control



Cron Jobs

Thank You
